

Real-Time Coronal Magnetic Field Modeling using Physics-Informed Fourier Neural Operators

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Abstract

Coronal magnetic fields are a driving force behind the formation and evolution of solar energetic particle events, which are known to disrupt critical ground and space infrastructure, causing significant economic damage. However, studying the Sun's corona has been a significant challenge in heliophysics due to the complexity of the magnetic field structures in solar active regions. Magnetohydrodynamics (MHD) simulations help us model these complex environments but require solving specific partial differential equations over a wide range of spatial and temporal scales, making their computation take days to weeks to solve. This study combines the divergence-free condition $\nabla \cdot \vec{B} = 0$ with Tensorized Fourier Neural Operators (TFNO) to predict the coronal vector magnetic field $\vec{B}(x, y, z)$ up to 8.3 Megameters (Mm) above the photosphere from a single 33 km tall layer $\vec{B}(x, y, 0)$. The model was trained on 254 high-resolution samples of the simulations, achieving $R^2 = 0.820$, $RE = 0.326$, and predictions within 2 seconds per sample. This work demonstrates the viability of a physics-informed approach for predicting non-force-free coronal magnetic fields and provides a faster method of studying them compared to simulations using Bifrost MHD.

Keywords: Coronal Magnetic Fields, Magnetohydrodynamics, Fourier Neural Operator

1. Introduction

Solar Flares and Coronal Mass Ejections (CMEs) are known to disrupt critical technologies on Earth, endanger astronauts, and damage satellites (1; 2; 3). These intense radiation storms can cause millions of dollars in damage within minutes (3). Predicting these events is central to minimizing their impact on Earth. Flares and CMEs originate from solar active regions consisting of intense photospheric and coronal magnetic fields (4; 1; 2). Current prediction models use magnetic field data from only the photosphere, the visible surface of the Sun, in active regions to forecast flares and CMEs (4; 1; 2). However, magnetic fields throughout the corona, the outermost layer of the Sun's atmosphere, are also crucial in the development and evolution of these energetic events (5; 6; 7; 8; 9). Computational simulations of these regions take weeks to months of compute time on supercomputers because of the structural complexity of the magnetic fields within the corona (5; 7). Due to this limitation, it is impractical to use coronal magnetic field data in machine learning models for real-time solar flare and CME forecasting.

The most common approaches to modeling coronal magnetic fields are Magnetohydrodynamic (MHD) simulations and Nonlinear Force-Free Field (NLFFF) approximations (5; 6; 7). NLFFF approximations assume a corona where the plasma- $\beta \ll 1$, indicating that the ratio of plasma pressure to magnetic pressure is very low and non-magnetic forces are negligible (6; 7). This approximation works well for static snapshots of coronal magnetic fields. However, even in low- β , NLFFF cannot model the evolution of these magnetic fields over time due to the neglect of nonmagnetic forces such as

gravity, pressure, heating, etc., that are not negligible under realistic conditions (6; 7). In contrast, MHD simulations require solving the full magnetohydrodynamic equations and allow for accurate time-evolution modeling of solar flares and CMEs by including nonmagnetic forces (10; 5).

The Bifrost MHD is an MHD simulation framework for solar atmospheres that provides high spatial and temporal resolution. Carlsson et al. produced a realistic, full-scale, MHD simulation of the outer solar atmosphere using Bifrost MHD (10). Furthermore, this framework is useful for validating extrapolations of coronal magnetic fields from other methods such as NLFFF (10; 5). This makes Bifrost MHD the state-of-the-art simulation method for studying magnetic field structures in the corona and improving the accuracy of solar flare and CME forecasting models by incorporating realistic coronal magnetic field data.

Machine learning models can greatly accelerate MHD and NLFFF simulations by predicting the solutions rather than numerically solving for them, allowing the real-time extrapolation of coronal MHD simulations (11; 5; 6; 12; 13; 7; 14). Common approaches include the use of primarily physics-driven or data-driven models to predict the solutions. (7). Baty and Vigon use physics-informed neural networks (PINNs) integrated with partial differential equations (PDEs) from coronal magnetic field physics to accelerate the simulation speed of coronal magnetic fields (7). These physics-informed approaches are suitable for simple initial conditions but fail for more complex environments and require retraining for each new set of parameters (11; 5; 7). The development of the data-

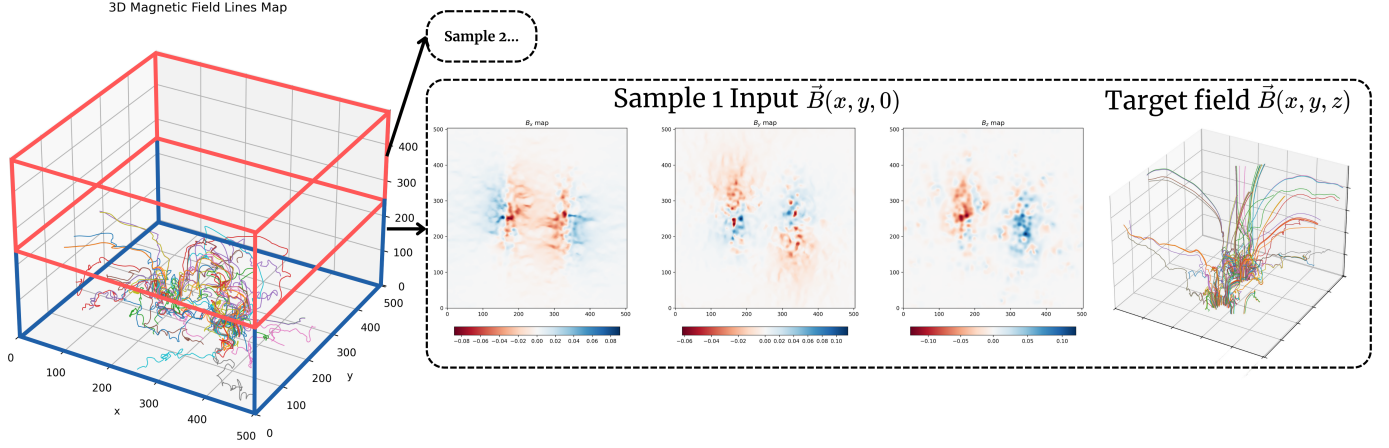


Figure 1: Input and target field generation

driven neural operator solves this issue (13; 15). These models learn the mapping between infinite-dimensional function spaces rather than between finite-dimensional vectors. In other words, unlike a neural network that maps input to outputs or numbers to numbers, neural operators generalize this by learning the mapping between entire input and output functions (15).

Yang et al. used a grid-constrained Fourier Neural Operator (FNO) to predict coronal magnetic fields based on NLFFF approximations (7). Their model achieved high accuracy compared to the simulations. Similarly, Jeon et al. used a model integrated with a custom physics-informed loss function to predict NLFFF simulations (6). This helped the model learn how magnetic fields behave rather than just learning the mapping between the conditions. However, combining physics-informed and data-driven approaches has only been used for NLFFF, and not MHD, simulations. Du et al. used FNOs to accelerate full-scale MHD models by training them on Bifrost MHD simulations (5). However, their model only predicts every 5th layer of the full 496-layer computational volume, resulting in 100 layers total. It learns the general structure of the magnetic fields, but not the fine-scale structures that would assist in forecasting solar weather and studying the corona. A better, but still realistic, number for this problem would be 247 consecutive layers, or half of the computational volume from the simulations. Jeon et al. showed how physics-informed loss functions help models generalize better (6). Using this approach for Bifrost MHD predictions would greatly increase accuracy and generalizability. Using a physics-informed loss function to help predict full-scale MHD simulations like Bifrost MHD and capture the fine, small-scale magnetic field structures has not yet been carried out.

This study aims to fill this gap by using a physics-informed Tensorized Fourier Neural Operator (TFNO) to predict 247 continuous layers of B-field, or coronal magnetic field, given a single input layer. Furthermore, this study diversifies the simulated dataset generated by Carlsson et al. (5) by using multiple initial conditions as described in Section 2.2, which

allows our models to better generalize to real active regions.

2. Methods

2.1. Review of Tensorized Fourier Neural Operators

Classical neural networks learn patterns from data to map inputs to outputs. The data these models use is represented using constant-size vectors (15). Neural operators generalize this to learning the mappings between infinite-dimensional input and output functions rather than fixed-dimensional vectors, allowing them to learn a family of PDEs on a continuous domain rather than just a single instance in a specific dimension, as PINNs do. Fourier neural operators learn in Fourier space, where complex spatial patterns can be better represented. In a layer of this model, a Fourier Layer, input data is transformed into frequency space using the Fast Fourier Transform. Only a given number of the most important frequency components, or modes, are retained in this layer. The retained modes are processed through multiple hidden layers before the data is transformed back into physical space (15). Tensorization organizes these large data structures into products of low-rank matrices, reducing the number of parameters in the model and increasing memory efficiency during training (5).

2.2. Data

This study used 314 high-resolution magnetic vector fields $\vec{B}(x, y, z)$ which have similar magnetic field topologies to those found on enhanced network areas on the Sun (10; 5). The magnetic vector field is defined such that:

$$\vec{B}(x, y, z) = \begin{cases} B_x(x, y, z) \\ B_y(x, y, z) \\ B_z(x, y, z) \end{cases} \quad (1)$$

where x, y, z are the dimensions of the computational volume. The original dataset was generated by Carlsson et al. and consisted of 157 magnetic vector field samples (10). To diversify this dataset, we split each volume in half, as seen in Figure

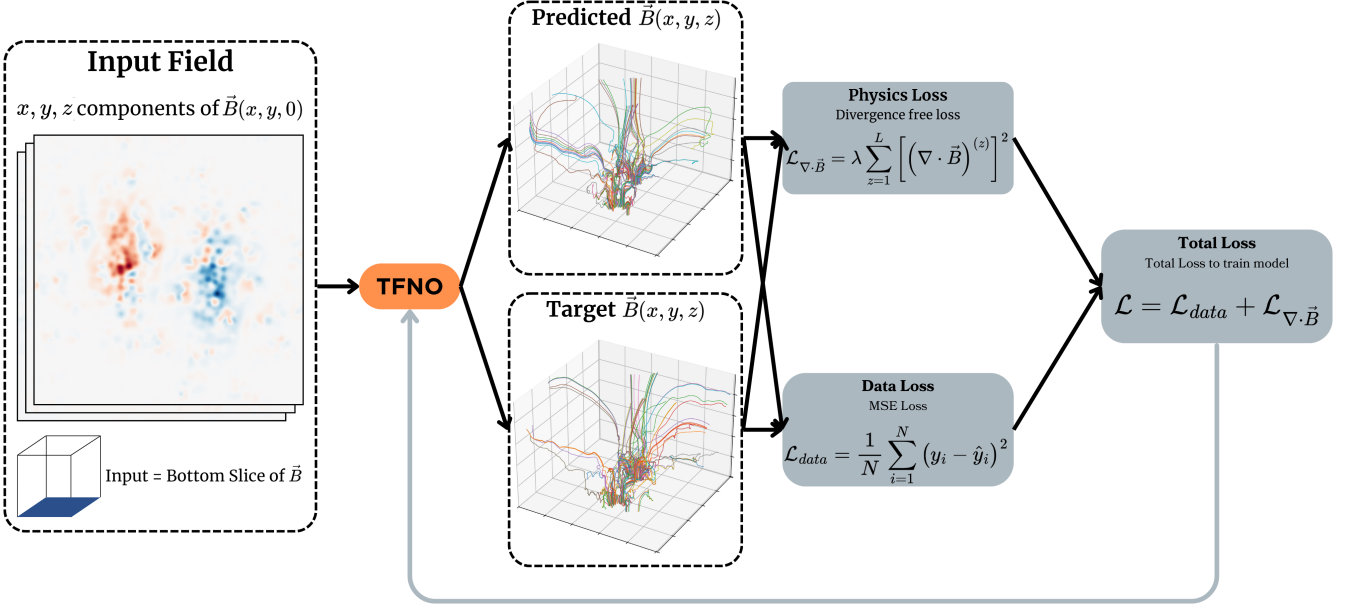


Figure 2: General structure of Physics-Informed Tensorized Fourier Neural Operator

1. This results in two separate vector fields, which we treat independently of each other. The goal of our model is to predict the vector field $\vec{B}(x, y, z)$ above the bottom layer $\vec{B}(x, y, 0)$ in each of these independent samples. By splitting each volume in half, we produced a total of 314 samples across multiple initial conditions, which helps our model generalize better to real active regions on the Sun. 254 samples were used during training, whereas 30 were used for validation, and the other 30 for testing. The size of each sample used in this study is $24 \text{ Mm} \times 24 \text{ Mm} \times 8.3 \text{ Mm}$, split into a grid of size $504 \times 504 \times 248$. The original data is available from the Hinode Science Data Center Europe in the en024048_hion dataset.

2.3. Physics-Informed Tensorized Fourier Neural Operator

We used a Physics-Informed Tensorized Fourier Neural Operator (TFNO) to extrapolate the target magnetic vector field $\vec{B}(x, y, z)$ using the bottom layer $\vec{B}(x, y, 0)$. The input vector field can be swapped for real photospheric vector field data from NASA's Solar Dynamics Observatory (SDO) to extrapolate B-fields for real active regions. The model incorporated a physics-infused loss term along with a data-driven loss function to help the model learn the behavior of the magnetic fields. The general structure of the model is shown in Figure 2.

2.3.1. TFNO structure

The structure of the TFNO was similar to that used by Du et al. for this task. The number of modes retained in the Fourier layer was set to (64, 64) with 64 hidden layers and a total of 6 Fourier layers. It took in 3 channels as input corresponding to the x, y, z components of the input vector field $B(x, y, 0)$. The model produced a total of 741 layers that are reshaped into a magnetic field cube with 247 layers, where each layer contains its own x, y, z components of the vector field. This corresponds to a magnetic vector field of height 8.3 Mm above the input

layer. We used a Tucker decomposition with rank 0.42 to tensorize the model (5). In the case of the Bifrost MHD models, the model outputs a cube of size $504 \times 504 \times 247$.

2.3.2. Physics-Informed Loss and Training

The magnetic fields predicted by our model must obey Gauss's law for the divergence-free condition, given by:

$$\nabla \cdot \vec{B} = 0 \quad (2)$$

We can use (2) to evaluate how well the predicted magnetic fields obey physical laws. By averaging it over each grid cell of each layer of the predicted field $\vec{B}(x, y, z)$, we can infer a loss term that helps the model learn how the magnetic fields behave during training. Using (2), we implemented the following loss term that describes the model's effectiveness:

$$\mathcal{L}_{\nabla \cdot \vec{B}} = \lambda \sum_{z=1}^L [(\nabla \cdot \vec{B})^{(z)}]^2 \quad (3)$$

where z represents the layer height and L represents the total number of layers. We also used mean squared error (MSE) in addition to the divergence-free loss. The $(\nabla \cdot \vec{B})^2$ value is relatively low compared to the MSE value, so we multiplied the divergence-free loss by an additional weight $\lambda = 10,000$. This ensures that both loss functions are equally represented during training. The total loss is calculated as the sum of the two loss functions. The structure of the loss calculations can be seen in Figure 2. The model was trained using the Adam optimizer and ReduceLROnPlateau Scheduler for a total of 200 epochs at a batch size of 1. The models trained by Du et al. predicted 100 layers, indexed every five, of the simulated B-fields and did not use a physics-informed loss function. The model in this study predicted the full B-field up to 8.3Mm. This makes the models

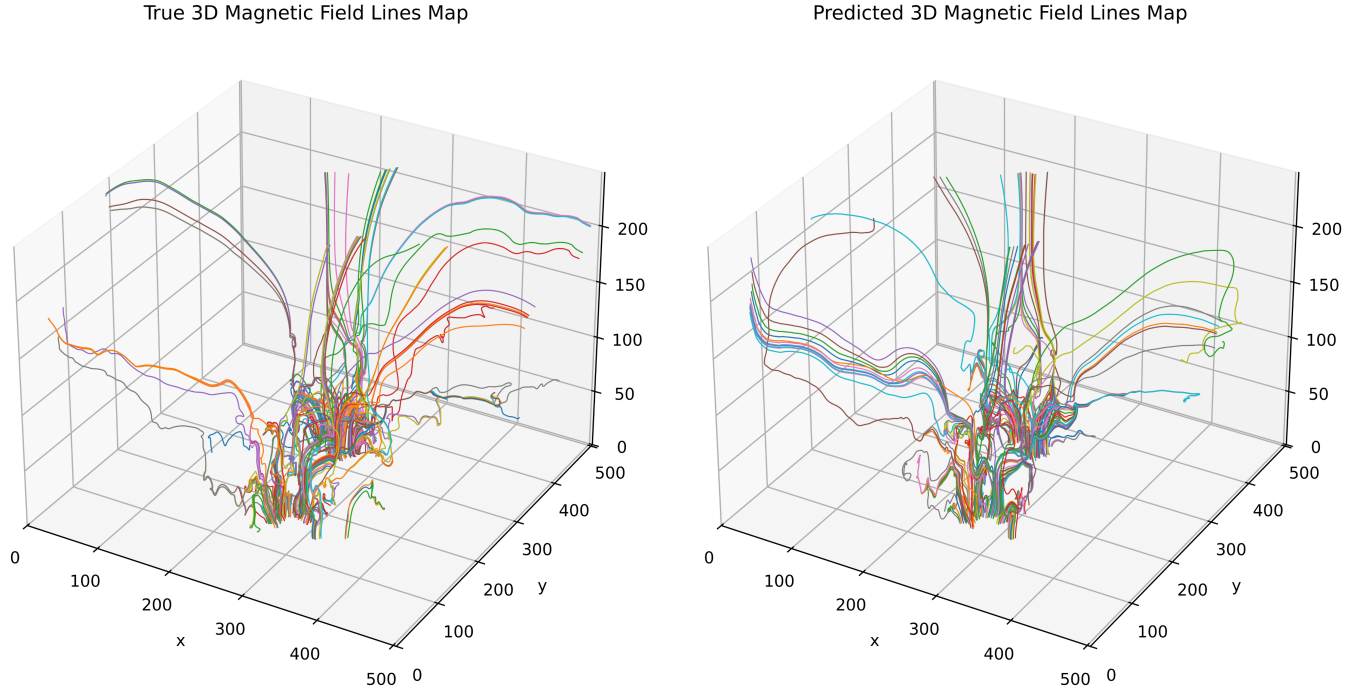


Figure 3: Ground Truth (left) vs Predicted (right) magnetic field lines for sample 60

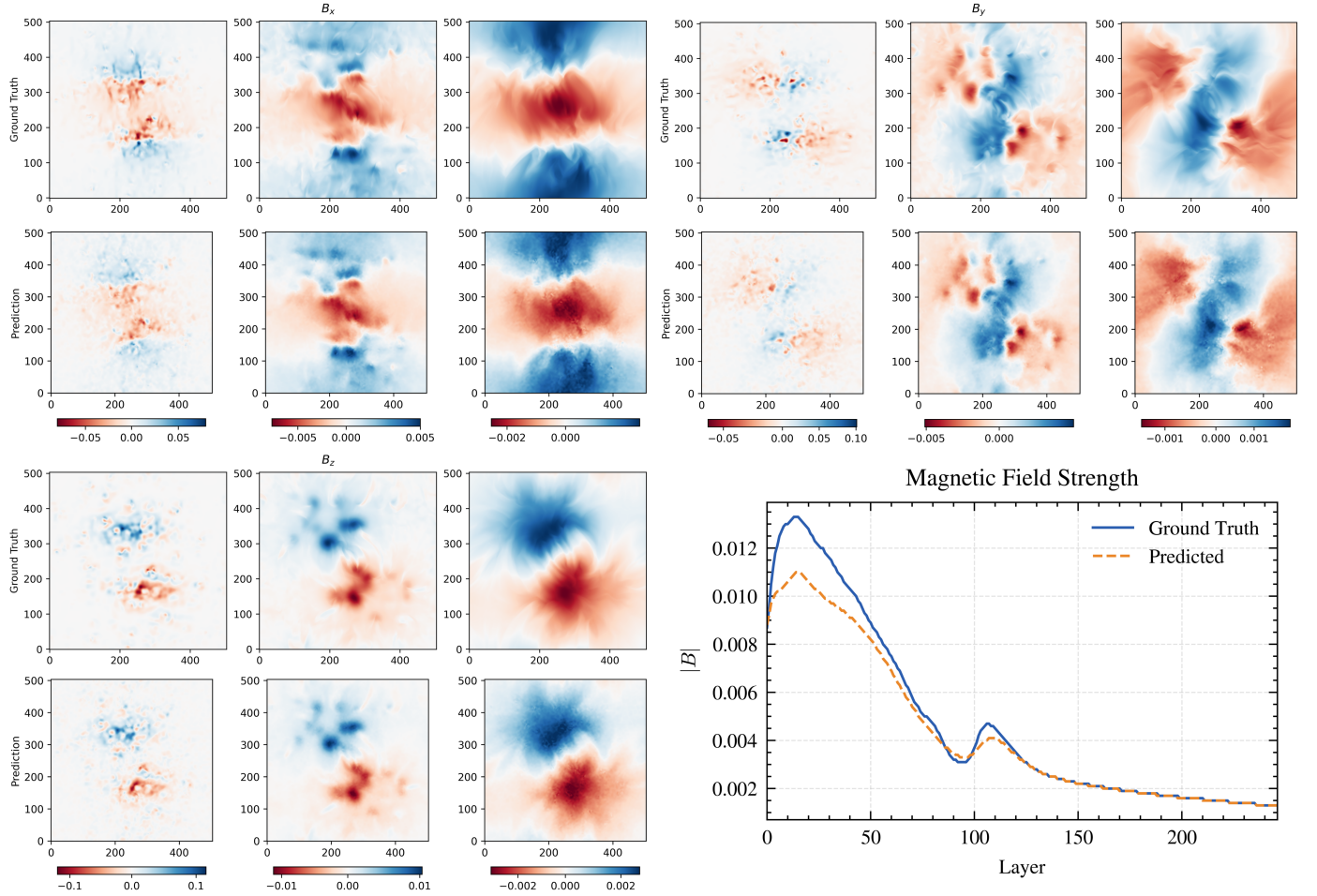


Figure 4: Ground Truth vs Predicted components of sample 60 at layers 2, 150, 248, and magnetic field strength

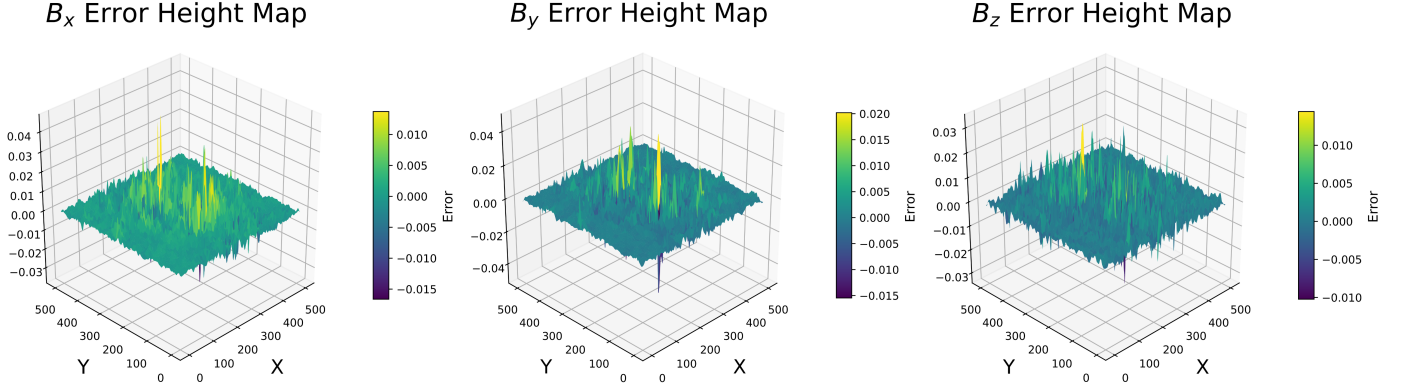


Figure 5: Error Height Map for Model Prediction of sample 60 at layer 2.



Predicted 3D Magnetic Field Lines Map

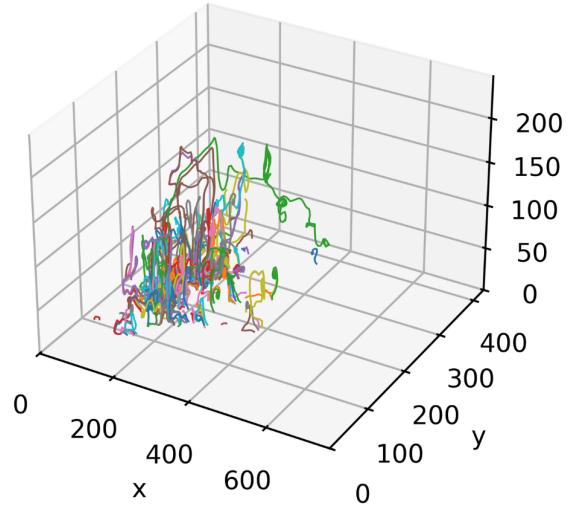


Figure 6: Full-disk solar image with automatically identified HARP regions (left) and the predicted coronal magnetic field for a real active region (HARPNUM 13294) on 1 June 2025, 00:00:00 TAI.

in this study approximately $2.5\times$ more complex and trained on more diverse data than the models trained by Du et al. (5).

2.4. Evaluation Metrics

To evaluate the effectiveness of the physics-informed TFNO, we calculated the correlation of determination:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (4)$$

as well as the mean relative error:

$$RE = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{y}_i - y_i|}{|y_i|} \quad (5)$$

Both of these metrics gave a comprehensive evaluation of the models and allowed for comparison to standard machine

learning models trained by other researchers for this task. The predicted magnetic field lines and layers of the predicted vector field $\vec{B}(x, y, z)$ were also compared to the simulated fields to serve as a qualitative measure for the effectiveness of the model. Finally, we plotted the mean magnetic field strength in each layer of the predicted $\vec{B}(x, y, z)$ against the simulated field strength to show the physical consistency of the predictions. Since the simulated magnetic fields are the target for our model, we will refer to them as the Ground Truth.

3. Results

Figure 3 shows the simulated magnetic field lines for sample 60 compared to the magnetic field lines predicted by the model. Table 1 shows the quantitative results of the model compared to the models trained by Du et al. We evaluated the models

using the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) as well; however, these metrics are not as useful because the model may learn to predict the mean magnetic field in each layer, allowing it to have low MAE and RMSE values without useful predictions. Due to this R^2 and RE serve as the best and main metrics for comparison.

Table 1
Numerical Results of TFNO Compared to Standard Machine Learning Models

Model	# Epochs	R^2	RE	MAE	RMSE
TFNO (247 layers)	200	0.820	0.326	0.0007	0.0032
CNN-RNN (100 layers)*	300	0.746	0.500	0.223	0.501
CNN-LSTM (100 layers)*	300	0.674	0.567	0.294	0.571

* The CNN-RNN and CNN-LSTM were trained by Du et al. [4] and only predict 100 layers instead of a full B-field like the TFNO trained in this study.

Figure 4 shows the ground truth vs. the model predictions for x, y, z components of sample 60 at various layers. The figure is split into 4 sections, where the top left, top right, and bottom left represent the x, y, z components of the vector field, respectively. In each section, the top row represents the ground truth, and the row below represents the model's predictions. The first, second, and third columns within each section represent layers 2, 150, and 248, respectively. The bottom-right section shows the magnetic field strength as a function of layer height in the simulated field of sample 60 compared to the predicted strength. The model approximates the field strength very closely, especially after 125 layers. Figure 5 shows the error maps for the x, y, z component predictions specifically at layer 2 of the sample.

4. Discussion

Figure 3 shows that the model accurately predicts the structure of the coronal magnetic field, compared with the simulated field on the left, which gives a visual representation of the accuracy of the TFNO. The model fails to fully predict the stray field lines that deviate from the main region of activity in the center. However, an isolated magnetic field line is less likely to interact with other stronger field lines, which makes it less important for flare predictions. The quantitative accuracy of the TFNO is further validated by the magnetic field strength plot in the bottom right of Figure 4, which shows that the predicted magnetic field strength converges to the ground truth as layer height increases.

An in-depth comparison of the simulated and predicted magnetic fields is provided in Figure 4 as well. The predicted components of the magnetic vector field at all layers closely resemble the simulated components. The error map (Figure 5) shows some regions of over/under estimates. However, much of the error map is shaded in the neutral green color, demonstrating low error. The error in predictions decreases as the distance between the input field and the predicted field increases. This can be seen through the convergence of the simulated and predicted magnetic field strength in Figure 4. The

physics-informed TFNO in this study achieved $R^2 = 0.820$ and $RE = 0.326$, which is higher than all of the standard machine learning models that only predict 100 layers (5), including the CNN-RNN and CNN-LSTM encoder-decoder models shown in Table 1. The TFNO also achieved a $MAE = 0.0007$ and $RMSE = 0.0032$, which are significantly lower than the CNN-RNN and CNN-LSTM trained by Du et al (5).

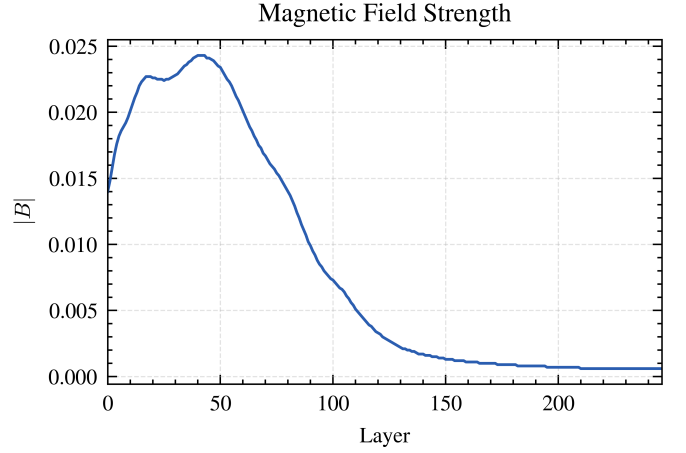


Figure 7: Magnetic Field Strength for Predicted $B(x, y, z)$ for HARPNUM 13294

Figure 6 shows the coronal magnetic fields predicted by the model for a real active region on the Sun. It predicted the magnetic fields above the active region with HARPNUM 13294 at midnight on June 1st, 2025. The magnetic field strength (Figure 7) features a mound-shaped peak and steadily decreasing strength following this peak, which is different from the field strength plot seen in the simulations produced by Carlsson et al (10). The model predicted these magnetic fields within 2 seconds, and the resulting data, such as the magnetic field strength, can be easily used to better predict solar flares and CMEs. The predicted active region was also of a different size than the training data; however, the use of neural operators extends the application of the model to regions of differing initial conditions and sizes, further validating its use for real-time coronal magnetic field modeling.

5. Conclusions

In this study, we developed a physics-informed approach for accelerating the modeling of full MHD coronal magnetic fields. We used a Tensorized Fourier Neural Operator, which incorporated a custom loss function enforcing the divergence-free condition $\nabla \cdot \vec{B} = 0$ to ensure the results would be physically consistent. The TFNO was trained on 254 samples of Bifrost MHD simulations and tested/validated on the remaining 60 samples. The results indicate that the model trained in this study was more effective than standard encoder-decoder machine learning models used for this task. It was used to model real coronal magnetic fields for an active region on the Sun in real time within 2 seconds. The model is able to accurately predict real

coronal magnetic fields for active regions of varying sizes due to the usage of neural operators, which learn input and output mappings of functions rather than a specific dataset. Our results validate the approach presented in this study as well as present a way for coronal magnetic fields to be used as input data for solar weather forecasting. Future work includes the use of additional simulations to better train the model and ensure the predictions are accurate. The use of more physics-informed loss functions, such as those modeling the curl of the magnetic field, could also be used to increase the effectiveness of the model.

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